
React Native



What is React Native ?

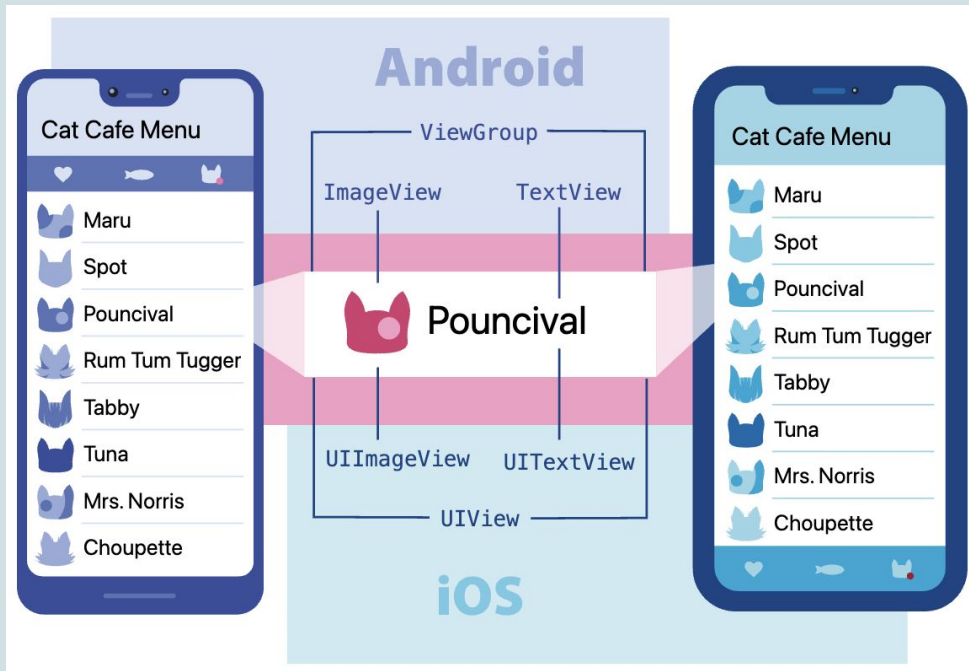
A JavaScript library for building native app user interfaces



If you know React
you already know React Native.



Cross-platform mobile development with React



Write once, run on iOS + Android + Web

Why React Native ?

You already know JavaScript → **start fast**

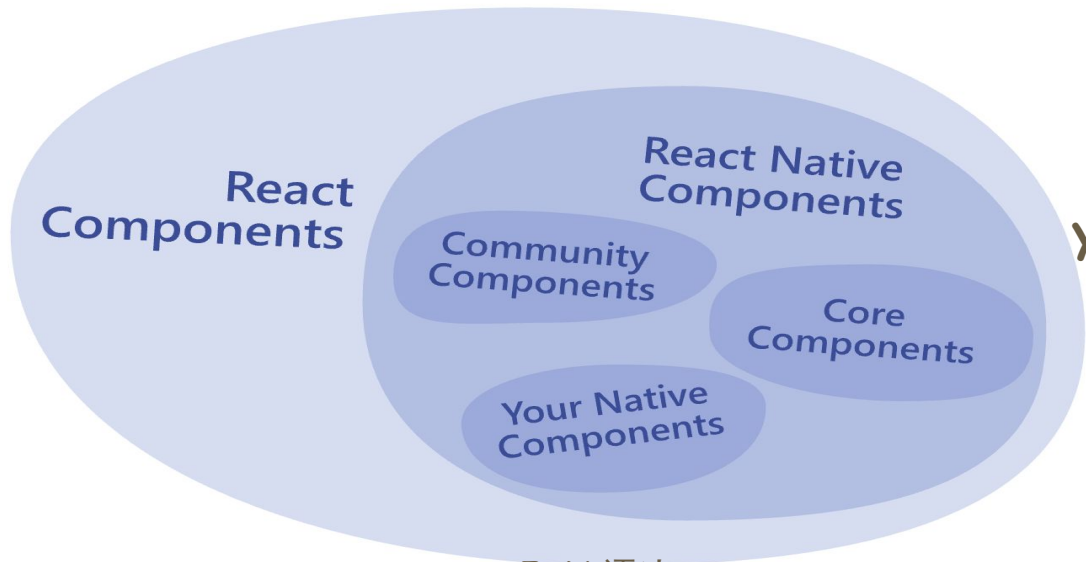
Few building blocks → **build apps quickly**

Cross-platform → **one codebase, many platforms**

Open source → **powered by the community**

React VS React Native ?

Same - React Native is built on React



If you know React,
you're already 90% there.

- ☐ JSX 语法
- ☐ useState / useEffect / useRef
- ☐ Props & Components
- ☐ TypeScript
- ☐ import / export



Different

	React	React Native
Target	Web Browser	iOS / Android
Container	div	View
Text	p / span	Text
Image	img	Image
Click	onClick	onPress
Style	CSS	StyleSheet
Font	@import	useFonts()
Routing	React Router	Expo Router



Toolchain



Step 1 — Install Node.js + npm

`npx create-expo-app@latest`

Step 2 — Install Android Studio

Set up Android Virtual Device (AVD)

Step 3 — Create Project

`npx create-expo-app my-app`

Step 4 — Run

`npx expo start`

→ Expo detects Android emulator automatically

```
a11111@wifi-172-23-68-174 s2 % npx create-expo-app@latest
```

Creating an Expo project using the `default` template.

To choose from all available templates (<https://github.com/expo/expo/tree/main/templates>) pass in the `--template` arg:

```
$ npx create-expo-app --template
```

To choose from all available examples (<https://github.com/expo/expo/tree/main/examples>) pass in the `--example` arg:

```
$ npx create-expo-app --example
```

```
? What is your app named? > my-app
```

What is Expo ?

Expo is more than a test tool

Framework

Routing

Native APIs

Build & Package

`npx create-expo-app`

Expo Go

Phone App

Scan QR Code

Real device test

One of many options

Testing Options



Way one — Expo Go → Scan QR, real device
same WiFi or tunnel

Way two — Android Studio → Virtual device (AVD)
npx expo start → auto launch

Way three — Web Browser → npx expo start → press W
fastest, not real mobile

Download Expo Go and scan the QR code to get started.



```
a11111@wifi-172-23-68-174 dog_app % npx expo start
Starting project at /Users/a11111/WorkSpace/react/example_practice/react_Native/dog_app
React Compiler enabled
Starting Metro Bundler
Recrawled this watch 1 time, most recently because:
MustScanSubDirs UserDroppedTo resolve, please review the information on
https://facebook.github.io/watchman/docs/troubleshooting.html#recrawl
To clear this warning, run:
`watchman watch-del '/Users/a11111/WorkSpace/react/example_practice' ; watchman watch-project '/Users/a11111/WorkSpace/react/example_practice'`
```



```
> Metro waiting on
exp://172.23.68.174:8081
```

```
> Scan the QR code
above with Expo Go
(Android) or the
Camera app (iOS)
```

```
> Web is waiting on
http://localhost:8081
```

```
> Using Expo Go
```

```
> Press s | switch to development build
```

```
> Press a | open Android
```

```
> Press i | open iOS simulator
```

```
> Press w | open web
```

```
> Press j | open debugger
```

```
> Press r | reload app
```

```
> Press m | toggle menu
```

```
> shift+m | more tools
```

```
> Press o | open project code in your editor
```

```
> Press ? | show all commands
```

```
Logs for your project will appear below. Press Ctrl+C
to exit.
```

Android Setup

Install Android Studio

Set up AVD (Android Virtual Device)

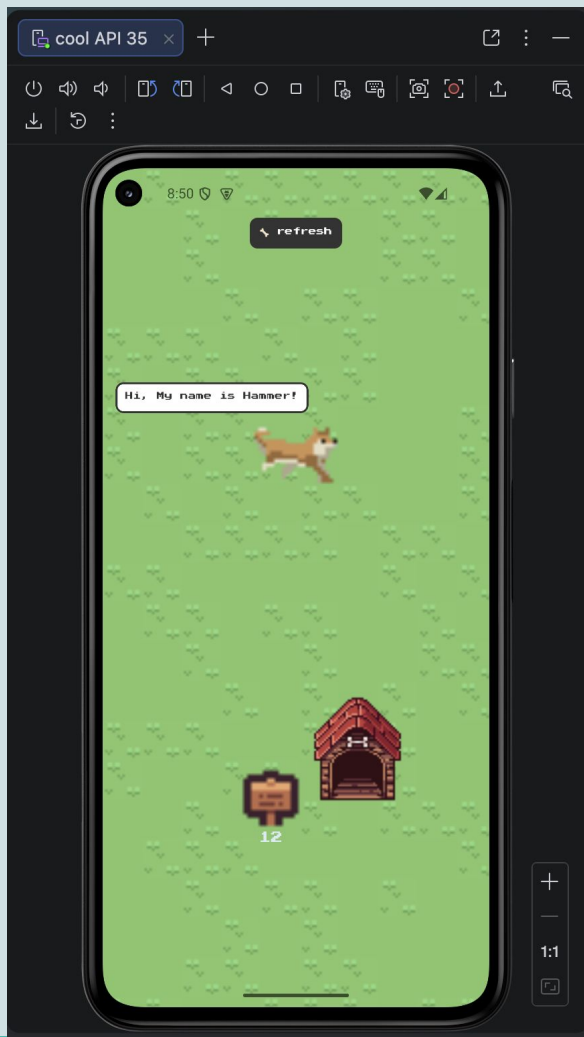
`cd file_name`

`npx expo start`

Expo auto-launches emulator



Source: reactnative.dev



Code Time...



First : Hello World

Second: compare versions